Fiscal Year:	FY 2022	**Task Last Updated:**	FY 05/27/2022
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Project Title:	Time and Space, Determining the Interactions Between the Circadian Clock and Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/26/2018	**End Date:**	01/04/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 01/04/2023 per NSSC information (Ed., 4/6/23) NOTE: End date changed to 7/25/2022 per NSSC information (Ed., 8/9/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Perera, Imara Ph.D. (North Carolina State University)		
Grant/Contract No.:	80NSSC18K1466		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The circadian clock, an internal timekeeping mechanism, enables organisms to temporally organize their molecular and biochemical activities so that they are optimally timed with environmental conditions. The circadian clock contributes to overall fitness in plants and animals. Our preliminary results support a connection between plant gravity sensing and responses and the circadian clock, which has not been explored/reported previously. We propose to examine this interaction in both directions in Arabidopsis plants: investigating the effects of the circadian clock on gravity perception and responses and the system-level effects of simulated microgravity on the molecular clock. In early spaceflight studies, the impacts of microgravity on the circadian clock and the subsequent effects on physiology in humans and other organisms were recognized, as well as the potential for profound implications for crew performance. In mammalian studies, there is evidence that microgravity and recovery from microgravity affect core components of the mammalian circadian clock. Yet a full examination of the molecular interplay between gravity and the circadian clock has not been reported in any organism. As with crew-performance, a disrupted circadian clock has a negative impact on plant growth and fitness. In plants, almost all aspects of growth, physiology, and environmental responses are modulated by the circadian clock. Therefore, understanding the interplay between changes in gravity, the circadian clock, and circadian-regulated output will have important implications for plant growth in future space missions. The experiments outlined here will identify not only the effects of changes in gravity levels on the core clock components but will also examine the genome-wide downstream effects on transcriptional timing. We observed that the time of day a gravitropic stimulus is provided affects the magnitude of the response. Furthermore, this response is altered in a circadian mutant. Based on these findings we hypothesize that, like other environmental stimuli, the response to gravity is gated by the circadian clock. We also examined transcriptomic datasets in GeneLab and found that core clock components are differentially expressed in response to gravity in multiple studies in Arabidopsis. This suggests that the circadian clock may be sensitive to changes in gravity. To examine how the interaction between the circadian clock and gravity is affected under simulated microgravity we will utilize the random positioning machine to evaluate the following hypotheses. Hypothesis 1: Gravity sensor(s) is/are controlled by the circadian clock. Therefore, the time of day the plants are transferred to microgravity will affect their response and the response to gravity will be altered in plants with mutations in the circadian clock. Objective: Determine the effects of the time of day and the circadian clock on gravitropic responses. Hypothesis 2: Gravity signals modulate the Arabidopsis circadian clock and circadian-regulated outputs. Therefore, clock components and/or clock outputs will be altered under simulated microgravity. Objective: Investigate how simulated microgravity affects the circadian clock and circadian-regulated outputs. This proposal focuses on the interaction between gravity and the circadian clock in a plant system, yet many principles of circadian circuits span across multiple organisms. Transcriptomic data will be generated and deposited into the GeneLab database. The system-level understanding of gravity perception and responses and the intersection with the circadian clock acquired here will contribute towards adapting plants to spaceflight and may also improve our understanding of the impacts of space travel on other organisms. Training of new generation US Space Biologists will focus on data analysis and integration of data sets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has two investigation tracks which have an impact on life on Earth. The first aspect is to understand how the time of day and the circadian clock affect the response to gravity in plants. The circadian clock controls many aspects of growth and development in plants and does this through daily regulation of transcription, translation, and other biochemical activities. Since most of the molecular components of the plant show a daily rhythm in their activities, the pathways and responses that depend on them also vary in efficiency throughout the day. We are using this underlying variation in biochemistry and physiology to uncover the pathways that plants use to sense and respond to changes in gravity. Understanding plant responses to gravity can have significant impacts on life on Earth as plant root growth, controlled by gravity, is an important factor in the ability of plants to access water. Our second investigation track focuses on the inverse connection--does gravity affect the circadian clock? The circadian clock of plants and animals enables organisms to optimize their response to the environment coordinating their internal molecular activities with externally recurring events such as sunrise and sunset or the changing of seasons. The internal circadian clock has been shown to enhance fitness in many species, and an altered clock can affect performance (e.g., yield in plants, response time in humans, and resistance to disease in both plants and animals). This molecular clock has developed in organisms within the confines of Earth with an ~24 h diel cycle and under ~1g of gravity. We are investigating if changes in gravity affect the functioning of the molecular clock, either through altering the oscillator itself or the output regulated by the circadian oscillator. If the clock is impacted by gravity, this can be used to understand the clock mechanisms. Insights into how the clock works and how it is connected to the downstream activities it regulates can be used to enhance the performance of crops, particularly as changing climate conditions allow us to grow crops in new environments. Two critical changes in plant growth are the poleward movement of cultivated areas and the expansion of growing seasons throughout the year. Both of these aspects involve components that are regulated by the circadian clock. Therefore, understanding how the clock functions can provide us with strategies for enhancing plant adaptability to new environmental conditions.
	The major goals of this project are to answer two fundamental questions: 1) Does the circadian clock affect the response to changes in gravity? 2) Does gravity affect the circadian clock? These questions remain unchanged, and this report covers and updates these questions and an additional question we initiated during COVID-related lab closures and travel prohibitions on the role of microgravity and telomere lengthening. As the lab and travel reopened, we shifted our focus back to the primary original two aims of the project asking about the interaction between the circadian clock and microgravity. The experiments related to Aim 1 (Does the circadian clock affect the response to changes in gravity?) were completed in prior reporting periods. During this period, the methods paper we submitted has been published and the manuscript concerning the findings has also been published. Briefly, the time of day and the circadian clock affect the gravitropic response, therefore the time of day experiments are performed is a critical feature that should be considered for gravitropic experiments. Additionally, we showed that it is likely not the statolith changes that drive this change in

Task Progress:	response. In this reporting period, we completed all experiments related to question 1 (Aim 1), prepared a review on the methods associated with appropriately accounting for time in microgravity experiments, and prepared a manuscript covering the findings related to question 1. These findings identify additional questions (which will not be answered by this proposal), including if changes in auxin levels or perception drive the changes in responses we observe. Additionally, we identified a major technical hurdle that needs to be resolved. Measurements of roots on agar plates in the random positioning machine (RPM) are limited by the two-dimensional approach that is commonly used -- seeds grown on the top of the media and images that are in two dimensions only. This approach could not capture the many roots we observed to grow into or out of the media when placed on the RPM. Identifying a method to measure the three-dimensional growth of roots on the RPM would improve our ability to identify how plants respond to microgravity in a more quantitative manner. This year we were able to travel to NASA Kennedy Space Center (KSC) to complete the experiments to answer the second question in Aim 2 (Does microgravity affect the circadian clock?). The plant growth apparatus designed by the team at KSC was tested in August and we were able to complete 24h of the 48h time course with two hour sampling of the plants. Unfortunately, (due to technical issues including power outages), we were not able to complete the full 48 hours in that trip. On a second trip, in November, the experimental issues were resolved for the RPM apparatus and growth conditions. However, we ran into issues with plant growth. In an attempt to reduce the amount of contaminating light the plants saw, I wrapped them too tightly and they overheated. The team at KSC allowed us to extend our stay and we were able to correct the issue and grow plants successfully. We completed the experiments and collected the tissue necessary for a full 48h time series of both control and RPM plants that will allow us to determine if the circadian clock is affected by growth on the RPM. When we returned with the samples, we extracted the RNA from all RNA samples, completing the extractions in February. We have optimized the library preparation and have begun library preparations for all samples. We anticipate completing the library preparations by the beginning to the middle of June. Some supply delays (pipette tips and some reagents) have slowed this process. While we have always been able to find the supplies, sometimes we must search for them and order them from new suppliers. Sometimes they are just on a short backorder. Combined across several supplies, these issues slowed down the process slightly. The library preparation has been improving. Initially, the student working on this had low yields or high adapter concentrations in the library. However, these have been resolved and the libraries are now at high concentrations with little to no adapters, and any remaining adapters can be removed in cleanup. In the previous report, we indicated an interest in evaluating amyloplast staining in circadian mutants and wild-type plants at multiple times of the day. We completed that experiment, and the results are included in the manuscript. There were no observed differences in staining at the different times of the day. However, unexpectedly, one circadian clock mutant showed significantly reduced staining of amyloplasts. This will be a project we will investigate in the future. Additionally, last period we reported evaluating telomeric length changes on some samples we previously collected on the RPM from KSC. In one experiment, we observed that telomere length was reduced on the RPM. However, we need to repeat this for confirmation, and it has been a lower priority now that the lab has opened back up and we can work on Aim 2 again.
Bibliography Type:	Description: (Last Updated: 05/25/2026)
Articles in Peer-reviewed Journals	Tolsma JS, Torres JJ, Richards JT, Perera IY, Doherty CJ. "Evaluating the effects of the circadian clock and time of day on plant gravitropic responses." Methods Mol Biol. 2022;2368:301-19. (Plant Gravitropism: Methods and Protocols. Noted as "in press" in prior report.). https://doi.org/10.1007/978-1-0716-1677-2_19 ; PMID: 34647263 , Oct-2022
Articles in Peer-reviewed Journals	Tolsma JS, Ryan KT, Torres JJ, Richards JT, Richardson Z, Land ES, Perera IY, Doherty CJ. "The circadian-clock regulates the Arabidopsis gravitropic response." Gravitational and Space Research. 2021 Dec 30;9(1):171-86. https://doi.org/10.2478/gsr-2021-0014 , Dec-2021
Dissertations and Theses	Tolsma JS. "Interactions between the circadian clock and gravity in Arabidopsis thaliana." Dissertation, North Carolina State University, June 2022. https://www.lib.ncsu.edu/resolver/1840.20/39840 , Jun-2022